6 The first term of a sequence of numbers is 18
The term-to-term rule for this sequence is "add 6"

(a) Is 603 a term of the sequence?
You must explain your answer.

No. 18 is even, and adding 6 always gives an even number.

603 is odd, so it cannot occur in this sequence.

1 mark
No, with parity reason

(1)

(b) Rizvi says,

"No terms of the sequence are multiples of 7"

Give an example to show Rizvi is wrong.

18, 24, 30, 36, 42, ...

42 = 7 × 6, so 42 is a counterexample.

1 mark
42 is in both sequences

(1)

(Total for Question 6 is 2 marks)